Angular HR/Employee Management System

Execution Guide for Presentation

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System Overview

This is a multi-role Employee Management System built with Angular that provides:

- **Role-Based Access Control** (Admin, HR, Manager, Employee)
- **Employee Directory Management** with CRUD operations
- **Leave Request Management** with approval workflow
- **Helpdesk Ticketing System**
- **Authentication & Session Management**
- **Chat Assistant** for quick actions
- **Material Design UI** with animations

Project Setup & Installation

Prerequisites

- **Node.js** (v18 or higher)
- **npm** (v9 or higher)
- **Angular CLI** (v17 or higher)

Installation Steps

Step 1: Navigate to Project Directory

```
cd c:\ANGULAR_PROJECT\FIRSTAPP
```

Step 2: Install Dependencies

```
npm install
```

This installs all required packages from package.json including Angular framework, Angular Material, RxJS, TypeScript, and development tools.

Step 3: Verify Installation

```
npm list
```

Execution Steps

Step 1: Start Development Server

Option A: Using npm (Recommended)

```
npm start
```

Option B: Using Angular CLI

```
ng serve
```

Expected Output:

```
⋮ Building...
✓ Compiled successfully.
✓ Build complete.

** Angular Live Development Server is listening on localhost:4200 **

Open your browser to http://localhost:4200/
```

Automatic Features:

- Application automatically opens in your default browser
- Hot reload enabled (changes auto-refresh)
- Compilation errors displayed in terminal and browser

Step 2: Access the Application

- **URL:** `http://localhost:4200/`
- **Default Route:** Login page (`/login`)

Step 3: Test Login Credentials

Admin Login

Email: `admin@company.com`

Password: `admin123`

Navigate to: `http://localhost:4200/admin`

HR Login

Email: `hr@company.com`

Password: `hr123`

Navigate to: `http://localhost:4200/hr`

Manager Login

Email: `manager@company.com`

Password: `manager123`

Navigate to: `http://localhost:4200/manager`

Employee Login

Email: `employee@company.com`

Password: `emp123`

Navigate to: `http://localhost:4200/employee`

Step 4: Run Tests

Run All Tests

```
npm test
```

Run Tests with Coverage

```
ng test --no-watch --code-coverage
```

> **Coverage Report Location:** `coverage/` directory

Step 5: Build for Production

```
ng build --configuration production
```

Output: Built files in `dist/` directory (production-optimized bundle)

Step 6: Stop Development Server

```
Ctrl + C
```

Spring Boot Backend Setup

This section covers the Spring Boot backend API server that powers the Angular frontend.

Backend Prerequisites

- **Java** (JDK 11 or higher)
- **Maven** (v3.6 or higher) OR **Gradle**
- **MySQL Database** (or other configured database)
- **Spring Boot** (v2.7+ or v3.0+)

Backend Installation Steps

> #### Step 1: Navigate to Backend Project Directory

```
cd path/to/your/spring-boot-backend
# Example:
cd c:\BACKEND_PROJECT\EMPLOYEE_API
```

Step 2: Configure Database Properties

Edit `application.properties` or `application.yml`:

```
spring.datasource.url=jdbc:mysql://localhost:3306/employee_db
spring.datasource.username=root
spring.datasource.password=your_password
spring.jpa.hibernate.ddl-auto=update
server.port=8080
```

Step 3: Build the Project

Using Maven:

```
mvn clean install
```

Using Gradle:

```
gradle clean build
```

Step 4: Run Spring Boot Application

Using Maven:

```
mvn spring-boot:run
```

Using Gradle:

```
gradle bootRun
```

Or run JAR file directly:

```
java -jar target/employee-api-1.0.0.jar
```

Expected Output:

```
  .   ____          _            __ _ _
 /\\ / ___'_ __ _ _(_)_ __  __ _ \ \ \ \
( ( )\___ | '_ | '_| | '_ \/ _` | \ \ \ \
 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
  '  |____| .__|_| |_|_| |_\__, | / / / /
 =========|_|==============|___/=/_/_/_/
 :: Spring Boot ::                (v3.0.0)

Started EmployeeApplication in 2.345 seconds
```

Backend API Endpoints

Method	Endpoint	Description
POST	`/api/auth/login`	User authentication
POST	`/api/auth/reset-password`	Reset user password
GET	`/api/employees`	Get all employees
POST	`/api/employees`	Create new employee
PUT	`/api/employees/{id}`	Update employee

DELETE	`/api/employees/{id}`	Delete employee
GET	`/api/leave-requests`	Get leave requests
POST	`/api/leave-requests`	Create leave request
POST	`/api/tickets`	Create support ticket
GET	`/api/tickets`	Get support tickets

Backend Server Information

> **API Base URL:** `http://localhost:8080`
> The Angular frontend communicates with this API on port 8080.

Running Frontend & Backend Together

Terminal 1 - Start Spring Boot Backend:

```
cd path/to/backend
mvn spring-boot:run
# or using gradle
gradle bootRun
```

Terminal 2 - Start Angular Frontend:

```
cd c:\ANGULAR_PROJECT\FIRSTAPP
npm start
```

> **Success Checklist:**
>
> - ✓ Backend running on http://localhost:8080
> - ✓ Frontend running on http://localhost:4200

- ✓ Database connected and initialized
- ✓ CORS configured in Spring Boot (if needed)
- ✓ Both servers accessible without errors

Spring Boot Troubleshooting

Port 8080 Already in Use

Edit application.properties:

```
server.port=8081
```

Then update Angular environment config to point to new port.

Database Connection Error

Verify connection properties in application.properties:

```
spring.datasource.url=jdbc:mysql://localhost:3306/employee_db
spring.datasource.username=root
spring.datasource.password=your_password
```

Build Errors

```
mvn clean install -DskipTests
# or
gradle clean build --no-test
```

Features by Role

Admin Role

Access: `http://localhost:4200/admin`

Full Employee Directory Add Employees Edit Employees Delete Employees

Reset Passwords Manage System Resources

HR Role

Access: `http://localhost:4200/hr`

Full Employee Directory Add Employees Edit Employees Delete Employees

View Leave Requests

Manager Role

Access: `http://localhost:4200/manager`

Employee Directory (Read-Only) View Pending Requests Approve/Reject Leaves

Team Management Cannot Add/Edit/Delete

Employee Role

Access: `http://localhost:4200/employee`

Personal Profile Apply for Leave View Leave History Create Tickets View Tickets

Chat Assistant

Architecture

Project Structure

```
FIRSTAPP/
├── src/
│   ├── app/
```

```
│   │   ├── admin/              → Admin dashboard
│   │   ├── auth/               → Authentication & guards
│   │   ├── chat/               → Chat assistant widget
│   │   ├── employee/           → Employee components
│   │   ├── employee-leave/     → Leave management
│   │   ├── employee-tickets/   → Ticketing system
│   │   ├── hr/                 → HR dashboard
│   │   ├── login/              → Login component
│   │   ├── manager/            → Manager dashboard
│   │   └── app.component.ts    → Root component
│   ├── environments/           → Environment configs
│   └── main.ts                 → Entry point
├── angular.json                → Angular config
└── package.json                → Dependencies
```

Key Components

Component	Purpose
LoginComponent	Authentication entry point
EmployeeComponent	Employee directory
ManagerLeaveApprovalsComponent	Leave approval workflow
EmployeeLeaveComponent	Leave request submission
ChatWidgetComponent	AI assistant interface

Technologies Used

Frontend Framework

- Angular 17+
- TypeScript

- RxJS (Reactive programming)
- Angular Material (UI components)

Backend Framework

- Spring Boot 2.7+ or 3.0+
- Java (JDK 11 or higher)
- Spring Data JPA (ORM)
- Spring Security (Authentication & Authorization)
- Spring REST API

Database

- MySQL (Recommended)
- PostgreSQL (Alternative)
- JPA/Hibernate for ORM

Build Tools

- **Frontend:** Angular CLI, Webpack, TypeScript Compiler
- **Backend:** Maven or Gradle

Testing

- **Frontend:** Jasmine (Testing framework), Karma (Test runner)
- **Backend:** JUnit, Mockito, Spring Boot Test

Quick Reference Commands

Command	Purpose
`npm install`	Install dependencies

`npm start`	Start dev server
`npm test`	Run unit tests
`ng build --prod`	Production build
`ng serve --port 3000`	Custom port
`ng lint`	Code linting
Spring Boot Backend Commands	
`mvn clean install`	Build Spring Boot project
`mvn spring-boot:run`	Run Spring Boot application
`gradle clean build`	Build with Gradle
`gradle bootRun`	Run with Gradle
`java -jar app.jar`	Run JAR file
`mvn test`	Run backend tests

Troubleshooting

Port 4200 Already in Use

```
ng serve --port 4201
```

Dependencies Issue

```
rm -r node_modules package-lock.json
npm install
```

Build Errors

```
ng clean
npm install
npm start
```

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